The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.5.2 (prototype)
Log entry date	10 November 2026
Status	The interface changes were deployed on 10 November 2026 by the merge of pull request #5 into the GitHub main branch (commit 1947993). The 0.5.2 documents and the version number follow on the branch docs-0.5.2.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch
Previous version	0.5.1, 5 November 2026, commit 694afcb (see Spaghetti_Engine_Dev_Log_v0.5.1.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.5.1 (5 November 2026) to version 0.5.2 of The Spaghetti Engine. Version 0.5.2 simplifies the analysis tabs for users who are not modelling experts: every tab opens with a short setup in numbered steps, shows its main result as a figure with a plain-language reading, and keeps the detailed tables, diagnostics, and method in fold-outs. It adds a metro map of the transition pathways and gives all tables and figures one style.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.5.2

Area	Change
Transition Pathways	Three-step setup with More options; a metro map with one color per intervention, dashed combined pathways, and sorting; one Pathway archetypes box; Advanced results; the lever profiles box removed.
Transition Point Analysis	Three-step setup; a prioritization matrix with tinted quadrants above the results table; one panel per outcome under Advanced results.
Sensitivity and Monte Carlo	The Sensitivity analyses group after Transition analyses; a three-step setup, a plain-language reading, and one curve panel per watched concept; a What these results mean box on Monte Carlo.
Baseline and Scenarios	One run card, a status line, and two new figures on Baseline Equilibrium; one block to choose and edit scenarios and a comparison with one panel per scenario.
Other tabs	A tiled System overview on Influence Metrics; feedback loops under the bridge concepts on Influence Routes; a shorter Model editor; relationship ends that follow the boxes on the Network tab.
One style	One style for all tables and figures; Method & diagnostics as a fold-out at the bottom of every analysis tab.
Record of 0.5.1	Version 0.5.1 was pushed as commit 694afcb on 5 November 2026.

3. The Changes in Detail

3.1 Transition Pathways

- **Setup.** Three steps (What should improve?, What can be changed?, How many stages may a pathway have?) and a More options fold-out with the desired levels, effort weights, construction settings, order constraints, expert sequences, and robustness test; its header names the settings changed from the defaults.

- **Metro map.** In the style of adaptation pathways maps (Haasnoot et al., 2013): each route a row along the cumulative intervention intensity, a route continuing another starting at its transition point with a vertical transfer line, one color per intervention, solid lines for one intervention and dashed lines in the stage's color for combined pathways, a filled circle at the end and a bar past a tipping point. A scorecard with the total outcome gain and each outcome's end state, outcome names in full. Routes are chosen as the top N by total outcome gain (or by the outcome sorted on), the selected pathway, and every shorter route they grow out of; they can be sorted as a network, by gain, by an outcome, or by intensity. Layout in src/pathwayMetroLayout.js, drawing in src/PathwayMetroMap.jsx.
- **Results.** One Pathway archetypes box; Advanced results with the effort-gain plane, the selected pathway, all pathways, the portfolios, and the invariance test; the lever profiles box removed (its CSV export remains); Method & diagnostics at the bottom.

3.2 Transition Point Analysis

- **Setup in three steps.** Prioritization matrix with four tinted quadrants (split at a transition point of 0.5 and at the mean absolute effect), numbered dots, and a list per quadrant, above the results table; flat outcomes are left out. Advanced results with one panel per outcome (response curve, marginal-benefit bars, and transition point; src/TransitionSmallMultiples.jsx), the KPI cards, and the overlay charts.

3.3 Sensitivity analyses and Monte Carlo

- **Sensitivity Analysis.** The group moved after Transition analyses. Three-step setup, a plain-language reading, one curve panel per watched concept, and a tornado chart in the common style; elasticity metrics, impact ranking, and response table under Advanced results.
- **Monte Carlo.** A What these results mean box with fixed rules on the 5 to 95 percent range (Annex, Section 11.4).

3.4 Baseline Equilibrium and Scenarios & Simulation

- **Baseline Equilibrium.** A Run the baseline card, a status line with summary tiles, How the system settles (one panel per module with gridlines, markers, and end labels), and Where each concept settles (bars by module in two columns, an open circle at the start); diagnostics, convergence table, and raw vectors under Advanced diagnostics. Figures in src/AnalysisFigures.jsx.
- **Scenarios & Simulation.** One Your scenarios block: scenarios as tabs with compare boxes, a plain-language list of each scenario's changes, and Add a change (push up, push down, hold at a value). A comparison with one panel per scenario, hatched bars for concepts held fixed, and a switch between change against the baseline and end state. Advanced results with the method, the end-state chart in the same style, the table, and the trajectories.

3.5 Other tabs and the common style

- **Influence Metrics.** A tiled System overview in three groups; multi-scale analysis and metrics by module under Advanced results; How these metrics are computed and Method & diagnostics below.
- **Influence Routes.** A setup card and summary tiles; the feedback loops directly under the bridge concepts and the routes between modules; Method & diagnostics at the bottom.
- **Model editor and Network.** A short introduction, the two lists, and an Import and export tables fold-out for the matrix import and the table export. A relationship end floats to the side of its box that faces the other concept, and is pinned only when dropped on a connection dot.

- **Common style.** One style for all tables and figures, and Method & diagnostics as a fold-out at the bottom of every analysis tab (MethodPanel with a fold option).

4. Implementation Choices

Topic	Choice in the tool	Reason
Figures drawn as SVG	The new figures are plain SVG components with their own viewBox, not Recharts.	Full control over labels, wrapping, and small multiples; the figure export scales a viewBox to its on-screen size, and the workbook capture works as before.
Which pathways the metro map shows	The top N routes, the selected one, and their shorter routes.	A map of every pathway would be unreadable; adding the shorter routes keeps the network connected.
Pinned relationship ends	Only an end dropped on a connection dot is pinned.	Ends set by the direction of drawing went stale when boxes moved; older models therefore float on opening.
Interpretation rules	Fixed thresholds on percentile ranges (Monte Carlo) and on the mean effect (matrix).	Readable summaries for non-experts; the thresholds are stated in the annex as conventions.

5. Effect on Existing Results

- No analysis result changes: the computations are those of 0.5.1. Only how results are shown and in which order the tabs appear has changed.
- Saved models open unchanged. Relationship ends that a model recorded from the direction of drawing now float, so arrows can attach to another side of a box than in 0.5.1.
- The lever profiles are no longer shown as a box; they remain in the CSV export and the analysis workbook.
- Version raised from 0.5.1 to 0.5.2 in package.json.

6. Defects Found and Fixed

- **Figure export of SVG figures.** Figures drawn in their own units were cropped on export; the export now scales a viewBox to its on-screen size.
- **Stale arrow anchors.** Arrows drawn from a dot stayed on that dot when the boxes moved, and looped around the box; ends now float unless pinned.

7. Verification

- **Tests:** tests/test_pathway_metro.mjs (new, 20 checks on the metro layout and sorting), tests/test_marginal_benefit_tp.cjs, tests/test_transition_direction.cjs, tests/test_pathway_blank_inputs.cjs, tests/test_pathway_expert_robustness.cjs, and tests/test_fcm_layout.mjs pass.
- **Interface:** every changed tab was run on the peat meadow example in the browser, including the pathway analysis (5 levers, 3 outcomes, 3 stages), a transition point sweep, a baseline run, and a comparison of three scenarios; the fold-outs open and the figure exports produce complete images.
- **Build:** the production build completes without errors.

8. Deployment Status

- The interface changes were deployed on 10 November 2026 by the merge of pull request #5 into main (commit 1947993); Netlify rebuilt the site automatically.
- The 0.5.2 version number, this log, the manual, and the annex follow on the branch docs-0.5.2, with the PDFs in public/docs replacing those of 0.5.1 and the manifest rewritten.
- **Build note:** npm run build fails in the local OneDrive folder because of the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.5.1 logs still apply.
- The metro map shows a selection of pathways, not all of them; the effort-gain plane under Advanced results shows every pathway.
- The interpretation boxes use fixed cut-offs; they summarize the numbers but do not replace them.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.5.2)	How to use every feature. Section 1.2 lists the changes of 0.5.2; Sections 4 and 6 to 13 describe the reworked tabs.
The Spaghetti Engine - Methodological Annex (version 0.5.2)	The methods; Section 11.4 adds the rules of the Monte Carlo reading and Section 12.3 the quadrants of the prioritization matrix.
Spaghetti_Engine_Dev_Log_v0.5.1	The 0.5.1 release: tablets and phones, presentation view, messages inside the tool.
This Development Log	What changed in 0.5.2 and the technical work behind it.

11. Next Steps

- Try the simplified tabs with workshop participants who are not modelling experts, and adjust the wording of the plain-language readings.
- Describe the expert-defined sequences and the robustness families of 0.4.4 in the manual and the annex.
- Continue with the roadmap: version 0.6.0 (modelling made easier), including the Game mode on its own branch.